**UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Ordinary Level**

# MARK SCHEME for the May/June 2008 question paper

## 4024 MATHEMATICS

**4024/01** Paper 1, maximum raw mark 80

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| | | | | | |
|---|---|---|---|---|---|
| **1** | **(a)** $\frac{1}{14}$ | 1 | | In both parts, if answers decimal, accept fractions in working | |
| | **(b)** $4\frac{2k}{3k}$ | 1 | 2 | After 0 + 0, answers 0.0714(0) to 0.07145 | |
| | | | | **and** {4.665 to 4.67(0) or $\frac{14k}{3k}$} | C1 |
| **2** | **(a)** 6.7 oe | 1 | | | |
| | **(b)** (0).051 oe | 1 | 2 | | |
| **3** | **(a)** –8 | 1 | | | |
| | **(b)** $\frac{x-2}{5}$ oe | 1 | 2 | Must use $x$ | |
| **4** | 100 or 120 | 2 | 2 | Answer with more sig figs which rounds to this or figs 1 or 12 | C1 |
| | | | | **or** at least 2 of 9, $2^2$ and 0.3 seen | M1 |
| **5** | **(a)** 160.27 | 1 | | Accept 160 or 160.3 | |
| | **(b)** 6820 | 1 | 2 | | |
| **6** | 26 (cm) | 2 | 2 | $\frac{10}{\sin\theta}$ or $\frac{10\times13}{5}$ or $\sqrt{\{10^2+24^2\}}$ seen | M1 |
| | | | | Accept $\frac{10(\sin 90)}{\sin 5/13}$ for M1 | |
| | | | **[12]** | | **[12]** |
| **7** | **(a)** $32x + 120$ ($cm^2$) oe | M1 | | Accept any equivalent seen anywhere | |
| | **(b)** 8 (cm) | 2 | 3 | Their **(a)** = 376 oe seen | M1 |
| | | | | [Their **(a)** must be linear in $x$] | |

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|---|---|---|---|---|---|
| **8** | **(a)** 1 | 1 | | | |
| | **(b)** $\frac{1}{81}$ | 1 | | Accept (0).0123(0) to (0).01235 | |
| | **(c)** 27 | 1 | 3 | Accept ±27 | |
| **9** | **(a)** (*BDC* =) 42 (°) | 1 | | Accept all answers on the diagram if not seen in answer or working space | |
| | **(b)** (*ABC* =) 90 (°) | 1 | | | |
| | **(c)** (*ACB* =) 48 (°) **or** 138 – their (b) ft | 1 | 3 | ft only allowed if $0° < ACB \leqslant 96°$ | |
| **10** | **(a)** $y = 4x^2$ **or** $y = kx^2$ with $k = 4$ seen anywhere | 2 | | $4x^2$ seen **or** $y = kx^2$ seen | M1 M1 |
| | **(b)** $\frac{3}{2}$ and $-\frac{3}{2}$ oe www cao | 1 | 3 | Both required | |
| | | | **[12]** | | **[12]** |
| **11** | **(a)** Equilateral triangle | 1 | | Accept either word alone | |
| | **(b)** Rectangle | 1 | | | |
| | **(c)** Kite | 1 | 3 | | |
| **12** | $x = 2$ **and** $y = –3$ | 3 | 3 | One correct **with supporting working** **or** correct method to eliminate one variable condoning one arithmetic slip [reaching such as $11x = k$, $kx = 22$, $11y = k$ or $ky = (–)33$] | C2 M1 |
| **13** | **(a)** 3.75 or $3\frac{3}{4}$ or $\frac{30}{8}$ ($m/s^2$) oe | 1 | | Accept –3.75 etc. | |
| | **(b)** 270 (m) | 2 | 3 | Correct method to find complete area under graph | M1 |
| | | | **[9]** | | **[9]** |

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**14** **(a)** 259 (g) 1

**(b)** 20 (%) 2 3 120, $16\frac{2}{3}$ or 16.6 to 16.7 oe C1

**or** fig ($\frac{5}{25}$ (× 100)) M1

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**15** **Final answer** $\frac{8-t}{(2t-1)(t+2)}$ oe 3 3 $8 - t$ **and** $(2t - 1)(t + 2)$ or $2t^2 + 3t - 2$ M2

seen in single fraction, not necessarily together
**or** $3(t + 2) - 2(2t - 1)$ seen in single fraction with quadratic denominator M1
[condoning missing brackets]

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**16** **(a)** Ruled line from (0, –2) to (2, –1) drawn 1 Allow tolerance of ½ small square

**(b)** Correct region shaded **and** labelled **R** 2√ 3 Accept shaded in or out if R correct
Allow √ft if line wrong if possible **or** Shaded in or out, without R C1
Allow C1 √ft if line wrong if possible
If no line, shading with R marked to right of C1
$x = 0$, below $2y = 4 - 3x$, not bounded by $y = 0$

**[9]** **[9]**

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**17** **(a)** (3, –6) 1

**(b)** –4 1 Accept equivalents, such as $\frac{-16}{4}$ or $\frac{16}{-4}$

**(c)** $y = -4x + 6$ or 3 term equivalent 2 4 3 term line of gradient their **(b) or** which passes through (1, 2) or (5, –14) C1
Accept $y = -4x + c$ with $c = 6$ seen
$-4x + 6$ alone C1

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**18** **(a)** $9.19 \times 10^7$ (km) 2 Accept 9(or 9.2) × $10^7$
Correct answer not in standard form C1
**or** 150 000 000 – 58 100 000
or $15 \times 10^7 - 5.81 \times 10^7$ seen M1

**(b)** (0).15, $\frac{15}{100}$ or $\frac{3}{20}$ (terametres) 2 4 $1.5 \times 10^8 \times 10^3 / 10^{12}$ seen M1

$1.5 \times 10^{-4}$ oe seen S C1

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|---|---|---|---|---|---|---|
| **19** | **(a)** **(i)** $5x(3x + 2)$ | 1 | | In both parts, ignore extra brackets, condone missing outside brackets | | |
| | **(ii)** $(t + 3)(t – 5)$ | 1 | | If only solutions, accept factors in working | | |
| | **(b)** (0).6 oe | 2 | 4 | $4x – 1.2 = 3x – 0.6$ oe, or better seen | | M1 |
| **20** | **(a)** $\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$ | 2 | | At least 2 correct elements | | C1 |
| | **(b)** $\frac{1}{5}\begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$ oe | 2 | 4 | $\frac{1}{5}$ soi **or** $\begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$ soi | | C1 |
| | | | **[16]** | | | **[16]** |
| **21** | **(a)** $(t) < –1.5, –1\frac{1}{2}$ or $-\frac{3}{2}$ oe (e.g. $-\frac{6}{4}$) | 2 | | $t > –1.5$ **or** $–1.5$ alone | | C1 |
| | | | | **or** $–6 > 4t$ oe or better seen | | M1 |
| | **(b)** 58 | 2 | 4 | At least two of +4, +36 and +18 seen | | M1 |
| **22** | **(a)** –39 | 1 | | | | |
| | **(b)** **(i)** $3n + 1$ or **any** equivalent | 1 | | | | |
| | **(ii)** **(a)** $\frac{19}{36}$ | 1 | | | | |
| | **(b)** $\frac{3n+1}{n^2}$ or $\frac{\text{their (b)(i)}}{n^2}$ | 1 | 4 | | | |

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**23** **(a)** **(i)** (1 :) 300 000 — 1

**(ii)** 30 (km) — 1 — Accept answers in range 29.5 to 30.5

**(b)** Correct triangle drawn with sides 7 (± 0.2) and 6 (±0.2) cm, arcs visible — 2
No arcs seen, sides in wrong order and/or sides ± 0.4 cm and/or *AC*, *BC* not joined — C1
If *C* on wrong side of *AB*, otherwise correct — C1

**(c)** Perpendicular bisector of *AB* drawn — 1 — Within 0.2 cm of centre of *AB* and within 2° of perp
Minimum length 5 cm

**(d)** *F* and *G* marked correctly ft√ — 1ft — 6 — Each to be 3 (±0.2) cm from *C* on attempt at perp bisector, even if *C* is below *AB*

**[14]** **[14]**

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**24** **(a)** **(i)** 2 — 1

**(ii)** 2.52, $2\frac{13}{25}$ or $2\frac{26}{50}$ www — 3
2.52 oe or 2.5 seen www
or such as $\frac{126}{50}$ seen — M2
**or** $\frac{(0\times10)+1\times11+2\times8+3\times3+...}{10+11+8+3+...}$ (condoning one error) — M1

**(iii)** $\frac{3k}{175k}$ oe — 2
$\frac{7}{50}\times\frac{6}{49}$ or $\frac{7}{50}\times\frac{6}{50}$ or better seen — M1

**(b)** Ignore any block diagrams and outside the range $5 \leqslant t \leqslant 11$
Plots at heights 21, 11, 13 and 5 joined with straight lines — 1 — (Accept 10½, 5½, 6½ and 2½)

Time axis scaled and plots at 5, 7, 9 and 11 — 1 — 8 — independent

**[8]** **[8]**

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